

Daily Geometry Drill #7

Sheet by Henry Koduthore

Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Loci of points · Apollonius circle $\frac{PA}{PB} = k$ · geometry-logic conditionals · flaw-spotting in synthetic proofs.

Attempt each question before checking answers. Time yourself. No calculators.

Section A Rapid Recognition

(≤15 s each)

Name the locus straight off; only the standard families occupy these questions.

- Write down the locus of points whose distance from the origin is 5.
- Write down the locus of points equidistant from the two points (0, 0) and (4, 0).
- The locus of P with $PA = 2PB$, where $A = (0, 0)$ and $B = (6, 0)$, is a circle. It crosses the segment AB at one point. Write down the x -coordinate of that point.
- Write down the locus of points that are always 3 units from the fixed line $y = 4$.
- The locus of points whose distance to the point (0, 1) equals their distance to the line $y = -1$ is a parabola. Write down its equation.
- The locus of points whose distance to the point (1, 2) is exactly 3 is a circle. Write down its equation.
- Write down the locus of the midpoint M of every radius OQ of the circle $x^2 + y^2 = 4$ as Q runs around the circle (with O the origin).
- The equation $x^2 + y^2 = 2x$ describes a circle. Write down its radius.
- Write down whether the statement “ $x^2 + y^2 = 4$ implies $x = \pm 2$ or $y = \pm 2$ ” is true or false.
- Write down the locus of points P with $PA = PB$, where $A = (1, 3)$ and $B = (5, 3)$. (Give its equation in x .)

Section B Manipulation Drills

(≤50 s each)

Square both sides to force Apollonius loci into circle form, or reason about necessity/sufficiency for the logic items.

- The locus of P with $PA = 3PB$, where $A = (0, 0)$ and $B = (8, 0)$, is a circle. What is its radius? *(after tmua.fyi Challenge Mock 1 Paper 1 Q14)*

- A) 9
B) 6
C) 3
D) 8
2. Locus of points equidistant from $(0, 0)$ and $(0, 4)$ is?
A) $x = 2$
B) $y = 2$
C) $y = 4$
D) $x = 4$
3. The points satisfying $x^2 + y^2 \leq 4$ form a filled disk. Its area is?
A) 2π
B) 4π
C) 8π
D) 16
4. The points $A = (0, 0)$ and $B = (4, 0)$ are fixed. The point P moves in the region $y > 0$ so that $\angle APB = 45^\circ$. What is the locus of P ?
A) an arc of the circle with centre $(2, 0)$ and radius 2
B) an arc of the circle with centre $(2, -2)$ and radius $2\sqrt{2}$
C) part of the line $x = 2$
D) an arc of the circle with centre $(2, 2)$ and radius $2\sqrt{2}$
5. A point is known to lie strictly inside the unit disk $x^2 + y^2 < 1$. Which of the following statements is *necessarily* true for such a point?
A) $x^2 + y^2 < \frac{1}{2}$
B) $x + y < 1$
C) $x^2 + y^2 < 2$
D) $x^2 + y^2 > 0$
6. The Apollonius locus $PA = 3PB$ with $A = (0, 0)$, $B = (2, 0)$: what is the x -coordinate of its right endpoint on the x -axis?
A) 3
B) 4
C) 6
D) 9

7. To prove a point P lies on the unit circle $x^2 + y^2 = 1$, which single condition is *sufficient* on its own?
- A) $x^2 + y^2 < 1$
 - B) $x = -\frac{1}{2}$ and $y = \frac{\sqrt{3}}{2}$
 - C) P is not the origin
 - D) $x + y = 1$
8. Which property is necessary, but not sufficient, for a quadrilateral to be a rectangle?
- A) all four angles are right angles
 - B) the diagonals are equal in length
 - C) the diagonals are perpendicular
 - D) the diagonals are equal in length and bisect each other
9. A circle has equation $x^2 + y^2 - 4x - 6y + 9 = 0$. Write down its centre.
10. Write down the radius of the circle $x^2 + y^2 - 4x + 6y + 9 = 0$.

Section C Substitution & Structure(≤ 75 s each)

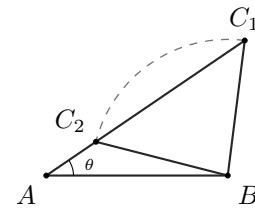
Hard synthesis: square the ratio for Apollonius, test necessary vs sufficient, and spot the hidden betweenness assumption. All Section C are MCQ.

1. The locus of P with $PA = 2PB$, where $A = (-2, 0)$ and $B = (4, 0)$, is a circle. What is its centre?
- A) $(6, 0)$
 - B) $(8, 0)$
 - C) $(10, 0)$
 - D) $(2, 0)$
2. The locus of points equidistant from the lines $L_1 : x + y = 0$ and $L_2 : x - y = 0$ is:
- A) $y = x$
 - B) $y = -x$
 - C) $y = |x|$
 - D) $x = 0$ or $y = 0$
3. *Claim.* If A , B and P lie on a circle with centre O , and P is on the major arc AB , then $\angle AOB = 2\angle APB$.

Proof. Draw the diameter PD . Triangles OPA and OPB are isosceles, so by the exterior angle theorem $\angle AOD = 2\angle APO$ and $\angle BOD = 2\angle BPO$. Adding, $\angle AOB = 2(\angle APO + \angle BPO) = 2\angle APB$.

Which statement about this argument is correct?

- A) It is correct and covers every position of P on the major arc.
- B) The exterior angle step fails, because $\angle OAP \neq \angle OPA$ in general.
- C) Adding assumes O lies inside $\angle APB$; when it does not, the angles must be subtracted, so that case is unproved.
- D) The claim itself is false when O lies outside triangle APB .
4. $PQRS$ is a quadrilateral. Consider the conditions: I. its diagonals bisect each other; II. its diagonals are equal in length; III. its diagonals are perpendicular. Which is the smallest set of these conditions that together guarantee $PQRS$ is a square? (after *TMUA 2020 Paper 2 Q7*)
- A) I and II
- B) I and III
- C) II and III
- D) I, II and III
- E) no set of them is enough
5. In $\triangle ABC$, $AB = 8$, $BC = 6$ and $\angle A = \theta$. Exactly two non-congruent triangles fit these data, and the larger has four times the area of the smaller. What is $\cos \theta$? (after *TMUA 2018 Paper 1 Q19*)
- A) $\frac{5\sqrt{7}}{16}$
- B) $\frac{3\sqrt{14}}{16}$
- C) $\frac{\sqrt{7}}{4}$
- D) $\frac{3}{4}$
6. The circles $C_1 : (x + 3)^2 + (y - 1)^2 = 25$ and $C_2 : (x - 8)^2 + (y - 1)^2 = 9$ have radii 5 and 3. How many common tangents do they have?
- A) 0
- B) 1
- C) 2
- D) 3
- E) 4
7. The point P moves so that $PA = 3PB$, where $A = (0, 0)$ and $B = (6, 0)$. What is the greatest possible value of PA ?
- A) 9
- B) 7.5
- C) 12
- D) 6



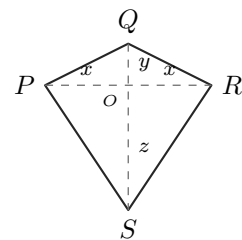
8. Fil finds the locus of P with $PA = 3PB$, where $A = (0,0)$ and $B = (2,0)$, by writing $(x-2)^2 + y^2 = 9$. What is wrong?
- A) Squaring loses sign information, so the locus is really a single point.
- B) The locus of $PA = 3PB$ is a straight line, not a circle.
- C) His equation says $PB = 3$; it never uses the distance to A .
- D) Nothing: $(x-2)^2 + y^2 = 9$ is the locus.

Section D Challenge Ramp

(≤90 s each)

Hardest 5: stack the week's tools — Apollonius, radical axes, area loci and geometric logic — under TMUA/BMO1 pressure.

1. The locus of P with $PA^2 + PB^2 = 34$, where $A = (-3,0)$ and $B = (3,0)$, is a circle. What is its radius?
- A) $2\sqrt{2}$
- B) 4
- C) $\sqrt{17}$
- D) $\sqrt{34}$
2. S is the square with vertices $(0,0)$, $(4,0)$, $(4,4)$ and $(0,4)$. What is the area of the region of points whose distance from the boundary of S is at most 1?
- A) $32 + \pi$
- B) $28 + 4\pi$
- C) $23 + \pi$
- D) $28 + \pi$
3. The diagram shows a kite $PQRS$ whose diagonals meet at O , with $OP = OR = x$, $OQ = y$ and $OS = z$. Which condition is necessary and sufficient for P , Q , R and S to lie on a circle?
(after TMUA 2022 Paper 2 Q11)
- A) $x = y = z$
- B) $2x = y + z$
- C) $x^2 = yz$
- D) $y = z$
- E) $y^2 = x^2 + z^2$
4. The circles $C_1 : x^2 + y^2 = 25$ and $C_2 : (x-4)^2 + y^2 = 9$ meet at P and Q . What is the length PQ ?
- A) 3
- B) 6
- C) $\frac{4\sqrt{14}}{3}$
- D) $\frac{24}{5}$



5. The points A and B are fixed, and $k > 0$ with $k \neq 1$. Prove that the locus of points P with $PA = k PB$ is a circle, and that this circle cuts every circle through A and B at right angles.

“A locus is a sentence the points all agree on; a geometric proof is only as strong as the condition you actually used.”

Found a mistake or want to contribute a sheet?

Visit the open-source project at github.com/The-CerealDev/speed-maths